CLASS 8 · CBSE · GANITA PRAKASH PART 1 · FORMULA BANK

Chapter 1: A Square and A Cube

Every formula and result of the chapter on one sheet

1	Square	$n^2 = n \times n$
2	Cube	$n^3 = n \times n \times n$
3	Units digits of a perfect square	0, 1, 4, 5, 6 or 9 only
4	Zeros at the end of a square	Always an even number of zeros
5	Sum of the first n odd numbers	$1 + 3 + 5 + \ldots + (2n - 1) = n^2$
6	The nth odd number	$2n - 1$
7	Next square	$(n + 1)^2 = n^2 + 2n + 1$
8	Numbers between n^2 and $(n + 1)^2$	$2n$
9	Number of factors	Odd only for perfect squares
10	Square root	If $y = x^2$ then $\sqrt{y} = x$ (positive root)
11	Cube root	If $y = x^3$ then $\sqrt[3]{y} = x$
12	Perfect square test	Prime factors form pairs
13	Perfect cube test	Prime factors form triplets
14	Hardy–Ramanujan number	$1729 = 1^3 + 12^3 = 9^3 + 10^3$